

Off-Campus Centre of Nitte (Deemed to be University)

Seventh Semester B. Tech (CBCS) Degree Examinations

December 2025

CC3313-1 – SOFTWARE DEFINED NETWORKS

(For CC)

Duration: 3 Hours

Max. Marks: 100

Note:

Part – A: Multiple Choice Questions: Answer all **Twenty** questions in the **OMR Sheet** provided. Each question carries equal marks.

Part – B: Descriptive Answer Questions: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II each** and **One full** question from **Unit – III**.

PART - A: MULTIPLE CHOICE QUESTIONS

20 Marks

1. The SDN controller communicates with network devices using _____.
A) Northbound APIs
B) Southbound APIs
C) East-West APIs
D) Middleware interfaces
2. The Control Plane in SDN handles _____.
A) Packet forwarding
B) Flow control and decision-making
C) Application hosting
D) Cable management
3. Which of the following is not a characteristic of SDN?
A) Centralized control
B) Vendor independence
C) Hardware-driven intelligence
D) Programmability
4. What is one major implication of SDN?
A) Hardware dependence
B) Simplified network management
C) Increased proprietary control
D) Lack of scalability
5. Which of the following best describes a distributed control plane?
A) All control logic is centralized
B) It uses static routes
C) No controllers exist
D) Control logic is shared among multiple controllers
6. Which component of SDN makes forwarding decisions?
A) Switch
B) Controller
C) Firewall
D) Router
7. The key characteristic of SDN switches is that they are _____.
A) Dumb forwarding devices
B) Intelligent routers
C) Firewalls
D) Servers
8. Which of the following is an alternate SDN method?
A) OpenFlow
B) NETCONF/YANG
C) SNMP
D) BGP
9. What is an OpenFlow "flow entry"?
A) A routing protocol
B) A data packet
C) A rule for forwarding packets
D) A hardware switch
10. Which version of OpenFlow introduced multiple flow tables?
A) 1.0
B) 1.1
C) 1.2
D) 1.3
11. Which of the following SDN characteristics improves scalability?
A) Distributed controllers
B) Hardware routing
C) Manual configuration
D) Fixed forwarding

12. Which field in a flow entry defines how packets should be treated?
 A) Match field B) Timeout field
 C) Cookie field D) Action field
13. The controller-to-controller communication uses _____ APIs.
 A) East-West B) Northbound
 C) Southbound D) Upstream
14. Which of the following best explains "alternate SDN methods"?
 A) Hardware-based routing B) Static control logic
 C) SDN frameworks not using OpenFlow D) Legacy network architectures
15. The abbreviation VXLAN stands for _____.
 A) Virtual Extensible LAN B) Virtual Exchange LAN
 C) Virtual Extended Line Access Network D) Variable Extended LAN
16. The main reason for using tunneling in data centers is to _____.
 A) Reduce physical cabling B) Create logical network overlays
 C) Increase power consumption D) Simplify VLAN removal
17. GRE stands for _____.
 A) Generic Routing Encapsulation B) General Routing Extension
 C) Gateway Routing Engine D) Global Router Exchange
18. The spine-leaf topology provides _____.
 A) High delay between switches B) Equal latency between any two devices
 C) Reduced redundancy D) Asymmetric links
19. Ethernet fabric eliminates the need for _____.
 A) Routing B) IP addressing
 C) Spanning Tree Protocol (STP) D) VLANs
20. Ethernet fabric is based on _____ topology.
 A) Mesh or spine-leaf B) Ring
 C) Star D) Linear Bus

PART - B: DESCRIPTIVE ANSWER QUESTIONS

Unit – I		Marks	BT*	CO*	PO*
1.	a) Define Software Defined Networks. Explain functionalities of data plane, with a diagram showing control and data plane of a network.	8	L*2	1	1
	b) Explain centralized network control approach through 4D proposal. Also, explain its three design principles and its core challenges, with an architectural design.	8	L2	1	1
2.	a) Explain general idea behind Forwarding and Control Element Separation (ForCES) in decoupling of forwarding and control planes. Describe its functional components, with design diagram	8	L3	1	1
	b) Explain Ethane with its three fundamental principles and architectural Diagram.	8	L2	1	1
3.	a) Explain LAN Emulation (LANE) network. Describe various role specific servers used to create a LANE.	8	L2	2	1
	b) Explain network orchestration with a diagram showing overview of the layout and operation of an orchestration management. Also mention its limitations.	8	L3	2	1
Unit – II					
4.	a) Explain SDN software switch and hardware switch, with respective switch anatomy.	8	L2	3	2
	b) Explain basic functions and its relationship to a controller in OpenFlow V.1.0 switch, with a neat diagram.	8	L3	3	2

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| 5. | a) | Explain alternate SDN method via hypervisor-based overlay networks, with suitable diagrams. | 8 | L2 | 3 | 2 |
| | b) | Explain Packet Matching process in OpenFlow 1.0. | 8 | L2 | 3 | 2 |
| 6. | a) | Explain the fundamental characteristics of SDN. | 8 | L2 | 4 | 2 |
| | b) | Explain how OpenFlow V.1.3 enforces parallelism by auxiliary connections? | 8 | L3 | 4 | 2 |

Unit – III

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| 7. | a) | Briefly explain the path technologies used in the in the Data Center. | 8 | L2 | 5 | 2 |
| | b) | Explain MAC-in-MAC (IEEE 802.1ah) and Q-in-Q (IEEE 802.1ad) encapsulation in Shortest Path Bridging standard. | 8 | L3 | 5 | 2 |
| 8. | a) | Explain Network Virtualization Using GRE (NVGRE) tunnelling technology with its packet format. | 8 | L2 | 5 | 2 |
| | b) | Explain how SDN via overlays, Open SDN, and SDN via APIs addresses Failure Recovery in data centers. | 8 | L3 | 5 | 2 |

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
